



Risky high protein diets

A recent study on mice showed that high-protein diets, such as Keto, boosts artery clogging and rupturing. <https://digit.in/feb20-11>



Decline in late cancer detection

A new research carried out in the US showed that there has been a decline in late-stage cancer detection. <https://digit.in/feb20-12>

Cancer is one of the top three killers in any country.”

One of the biggest challenges to delivering targeted therapies and treatments locally, is the lack of local data. Dr Deshpande says, “the data that we have in TCGA right now is collected in the US. It is regarding the US population. In India, we need to have this for our population, which we haven’t done before. What we are trying to initiate is, can we collect data like this for India, with Indian patients, so that over a period of time, over the next five years or so, we would have better data about our cancer, unlike their cancer. If we have that data about our cancers, then our treatments would be more appropriate for us, and we would be able to personalise it to individuals here.”

The genetics of the caucasian US population is different from South Asians and Indians. While certain treatments may work for caucasians, it does not necessarily mean that the same approach will work for South Asians as well. There is plenty of genetic variety within India as well. Dr Deshpande explains, “even within India, we have many pockets of DNA signatures for our population. It is very important that we do this in India. One of the things we are looking at is triple negative breast cancer. Breast cancer in India is different from breast cancer anywhere else in the world. Our breast cancer tends to afflict younger women, unlike the US or other places. Some of the diagnostic techniques such as mammograms don’t necessarily work very well. The disease is very deadly and is harder to cure. We have been able to find some DNA markers that are very specific to triple negative breast cancer.”

Dr Deshpande is leading the charge for getting a TCGA like setup in India. Although researchers in India have access to the TCGA and use the data, the data itself is American patients, and not local patients. Towards this end, Persistent Systems along with IISER (Indian Institute of Science Education and Research) and Prashanti

organised a TCGA conference in Pune in September 2019. Researchers presented how they used the TCGA, which was followed by a three day workshop where students were made familiar with the tools and techniques needed to handle such large databases.

Dr CB Koppiker, Breast Onco-surgeon, and Medical Director at the Prashanti Cancer Care Mission said, “with the declining cost of genomic technology, molecular/precision oncology is fast becoming mainstream in the cancer management in India. In my breast onco-surgery practice, I have seen the significant impact of tumour molecular profiling on better surgical and clinical management decision making. However, Indian cancers have uniquely different molecular attributes than their western counterparts. Hence, taking motivation from the TCGA project, it is timely and pertinent that the onco-community joins hands to initiate high impact, large scale, multi-disciplinary initiatives that are relevant to the Indian context. Prashanti Cancer Care Mission is committed to playing a proactive role in this effort.” A second conference is planned in September 2020, and the work on the database is expected to be underway by then.

Dr Deshpande says, “when this conference was going on, we pushed the medical guys to say, you know it is great to have a conference, but can we not do something more than that, and build a TCGA equivalent in India? I’ve been trying to spearhead this activity from our side to see if we can find funding and other resources across the country, which would support this kind of initiative. Persistent is quite confident about being able to handle the tech part of it, but this is not just a tech project, it needs cancer doctors, DNA researchers, and biologists to work together. I’ve been trying to speak to these groups and get them to cooperate and collaborate, and hopefully in the next three months we should be able to launch a TCGA equivalent here. It will be like the TCGA they have in the US, but it

will be for Indian patients, with data coming from here, stored in India, processed in India, and made available to whoever wants to do research across the world.” Dr Jean Claude Zenklusen, Director of TCGA is appreciative of the effort, saying “it is very encouraging that there is a willingness of performing large-scale genomics programs locally in India. I hope the results will be widely and openly distributed as the TCGA data was, and not restricted to commercial use.” Dr LS Shashidhara, Professor at IISER explains the need for such a platform, “considering human and microbiome diversity in India and our distinctive identities compared to the rest of the world, it is imperative that the more we share clinical data, the better we can manage cancer at the clinical level and thereby reducing health burden in the society. TCGA India initiative provides an opportunity for Indian researchers to generate quality data and share them freely on a single platform. This initiative will also impact South-East Asia, given our shared biological history.”

The international data sharing aspect of such a database can be beneficial to Indian cancer patients, just in the way local research based on the TCGA has benefited patients in the US. Dr Deshpande says, “It is important that when people are doing research on this topic, they include us as part of the data sample. If new drugs are going to be invented somewhere else in the world, they will be for our patients as well.” It can also help patients of Indian-origin elsewhere in the world. Prof Sunil Badve of the Indiana School of Medicine in the US says, “to make any headway against a complex disease like cancer, one has to use all the tools one can garner to understand its strength and weaknesses. Having large databases that catalogue all aspects of cancer is critical in this endeavour. TCGA-India could provide such a library and make a difference to the management of cancer patients in India and Indian diaspora which together amounts to approximately 20 percent of the world population.”